CORTEX AI

Multimodal Brain Tumor Clinical Decision Support System

Final Project Proposal

March 2026

1. Project Description

This project aims to develop an Artificial Intelligence–based clinical decision support system for brain tumor analysis using multimodal data.

The system will be evaluated using publicly available medical imaging datasets such as the BraTS dataset for brain tumor segmentation.

The system analyzes brain MRI scans using deep learning–based segmentation models to identify tumor regions and extract quantitative features. In parallel, radiology reports are processed using Natural Language Processing techniques to extract clinically relevant information and generate text embeddings.

Both imaging and textual features are integrated using a multimodal learning framework to improve tumor classification and diagnostic support. The system also provides interpretable outputs using explainable AI techniques such as Grad-CAM and SHAP to enhance transparency and trust in medical AI systems.

System Components

Cortex AI is built around five modules:

- **Computer Vision Module:** performs brain tumor segmentation from MRI scans using deep learning models such as U-Net or nnU-Net implemented in PyTorch. The module produces tumor segmentation masks and extracts basic quantitative features such as tumor size and location.
- **Natural Language Processing Module:** analyzes radiology reports using BioBERT / ClinicalBERT (HuggingFace Transformers) to extract clinically relevant information and generate text embeddings representing the radiology report.
- **Multimodal Fusion Module:** integrates image features and text embeddings using a Multilayer Perceptron (MLP) model. The system combines both modalities to produce tumor classification and diagnostic support indicators.
- **Explainability Layer:** provides interpretable results using Grad-CAM for MRI visualization and SHAP analysis for feature importance.
- **User Interface Module:** provides a platform for interacting with the system. It allows users to upload brain MRI scans and optional radiology reports, run the AI analysis pipeline, and view the

system's outputs including tumor segmentation results, classification predictions, and visual explanations.

2. Group Members & Roles

Name	ID	Email	Role
Nour Hossam (Team Leader)	21124765	nourhossamabdalrzak@gmail.com	NLP development
Ahmed Hossam	21054505	ahmedhossamfoda@gmail.com	Explainability layer
Ammar Kamal	21143792	ajka4142@gmail.com	Multimodal Fushion
Mariam Mohamed	21132570	mariammohamed111100@gmail.com	CV Development
Ibrahim Mahmoud	21055626	ibrahimmahmoudd4567@gmail.com	System Integration, Deployment

3. Objectives

The main goals of Cortex AI are:

- Develop a deep learning pipeline for automated brain tumor segmentation from MRI scans.
- Build an NLP module capable of extracting clinically relevant information from free-text radiology reports.
- Design and implement a multimodal fusion architecture that jointly learns from imaging and textual data to improve prediction accuracy.
- Produce interpretable AI outputs using Grad-CAM and SHAP.
- Validate the system on benchmark datasets (BraTS) and evaluate against clinical standard metrics.
- Deliver a functional prototype with a simple user interface for demonstrating the system workflow.

4. Tools &Technologies

Category	Tools / Frameworks
Deep Learning	PyTorch, MONAI
Machine Learning	Scikit-learn
NLP	HuggingFace Transformers, BioBERT, ClinicalBERT
Interface	Streamlit, Flask
Explainability	Grad-CAM, SHAP
Dataset	BraTS (Brain Tumor Segmentation Challenge)

Category	Tools / Frameworks
Data Processing	NumPy, Pandas
Visualization	Matplotlib, Seaborn
Version Control	Git, GitHub
Experiment Tracking	MLflow / Weights&Biases

5. Milestones & Deadlines

#	Phase	Weeks	Tasks
1	**Data Preparation**	Week 1-2	MRI preprocessing, dataset exploration, environment setup
2	**Computer Vision Development**	Week 2-3	Train MRI segmentation model
3	**NLP Module Development**	Week 4-5	Implement BioBERT text processing
4	**Multimodal Fusion**	Week 6-7	Combine image and text features
5	**Explainability Layer**	Week 8	Implement Grad-CAM and SHAP
6	**System Integration**	Week 9	Integrate the entire system
7	**Interface Development**	Week 10	Build Streamlit prototype
8	**Documentation**	Week 11	Presentation and Report preparation

6. Key Performance Indicators

The following metrics define the success criteria for Cortex AI:

6.1 Data Quality

Metric	Target Value
Percentage of missing values handled	**≥ 95%**
Data accuracy after preprocessing	**≥ 98%**
Dataset diversity (category representation)	**≥ 90%**

6.2 Model Performance

Metric	Target Value
Model accuracy / F1-Score	**≥ 90%**
Model prediction speed (Latency)	**≤ 500 ms**
Error rate (False Positive / False Negative Rate)	**≤ 10%**

6.3 Business Impact & Practical Use

Metric	Target Value
Reduction in manual radiological effort	**≥ 40%**
Expected cost savings in diagnostic workflow	**≥ 20%**
User satisfaction (clinician evaluation)	**≥ 80%**